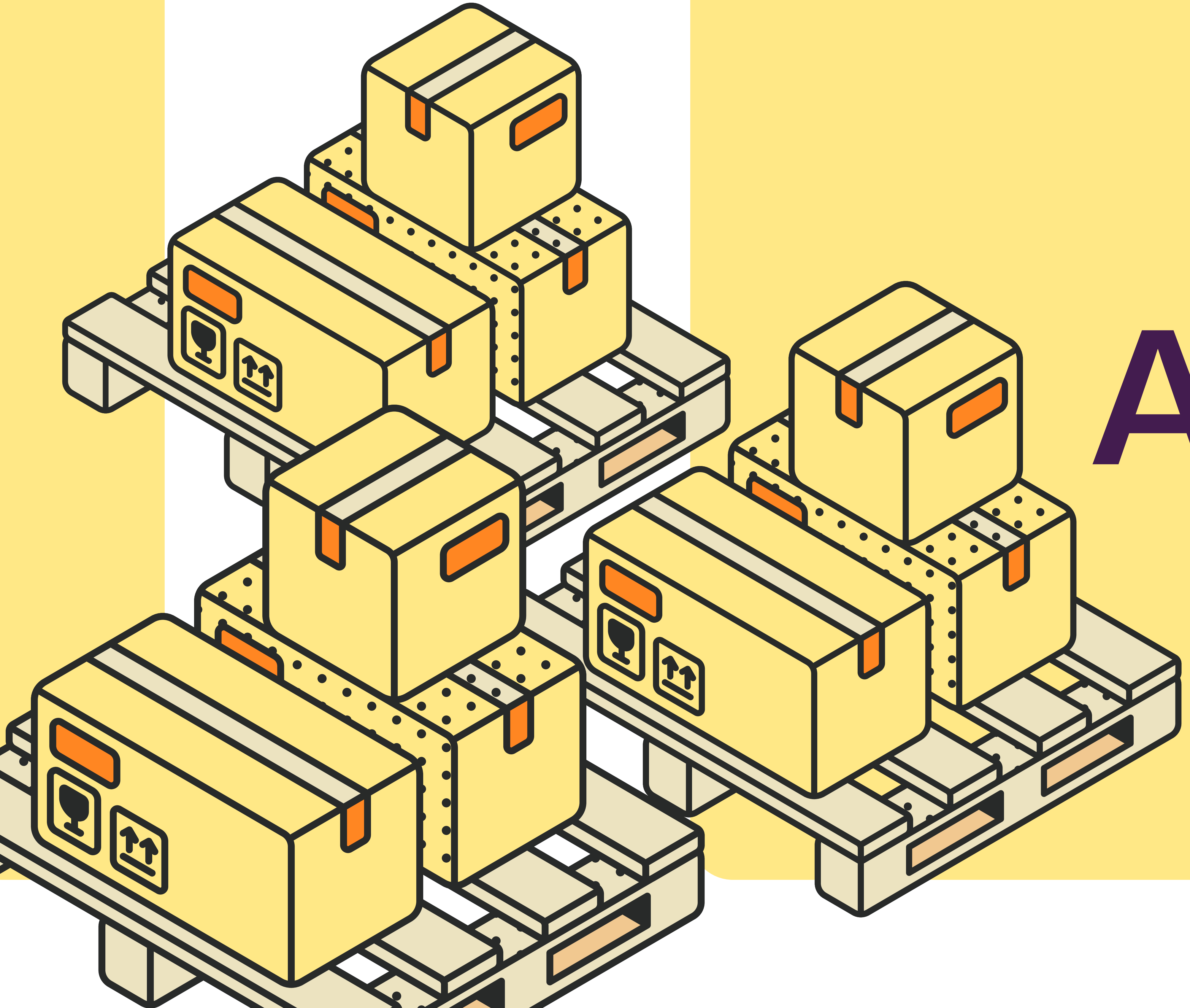

SUPPLY CHAIN ANALYTICS

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Why this dashboard is needed?

- Procurement and logistics teams are working with multiple operational tables: purchase orders, supplier master data, quantity logs, shipment records, carrier data, warehouse data, and monthly targets.
- Management cannot quickly compare actual spend against monthly targets.
- Supplier performance is split across pricing, quality, delivery, and risk fields.
- Logistics teams need carrier, transport mode, freight cost, and delay visibility.

Business Problem

• Spend Visibility - Compare actual procurement spend with monthly targets.
• Supplier Pricing Control - Identify unfavorable movements in actual prices versus contracted prices.
• Delivery Reliability - Measure on-time delivery and identify delayed shipments and root causes.
• Quality Leakage - Monitor rejected quantities and supplier quality performance.
• Logistics Cost Pressure - Analyze freight costs across carriers and transport modes.
• Executive Clarity - Provide decision-makers with KPIs, trends, and variance drivers in one view.

Business Questions

1. How does actual procurement spend compare with the monthly target?
2. Which product categories contribute most to spend variance?
3. Which suppliers demonstrate stronger or weaker pricing, quality, and delivery performance?
4. Where are the major Purchase Price Variance (PPV) drivers?
5. What is the overall on-time delivery performance, and which carriers or transport modes contribute to delays?

Business Questions

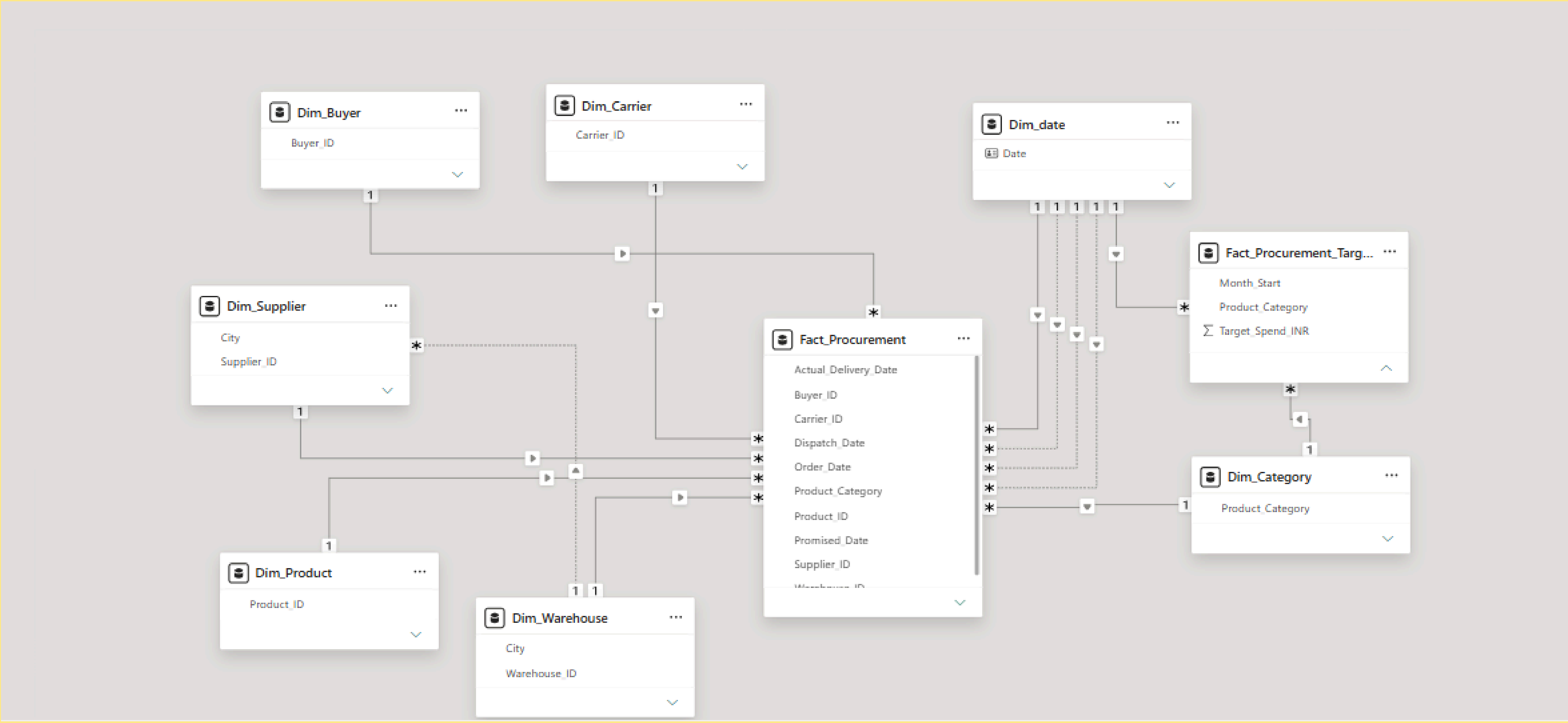
6. Where is quality leakage occurring through rejected quantities?
7. How does freight cost vary across carriers and transport modes?
8. Which suppliers present greater operational exposure or risk?
9. How do procurement and logistics KPIs trend over time?

Dataset & Data Model

The analytical model incorporates procurement, supplier, shipment, logistics, and target information. The model uses a Galaxy Schema with two fact tables: actual procurement data at PO-line grain and targets at category-month grain. Conformed Date and Category dimensions filter both facts, enabling accurate actualversus-target analysis.

An inactive delivery-date relationship is used where required for delivery-month analysis, with USERELATIONSHIP supporting accurate monthly OTD trends. The model deliberately avoids merging monthly targets into PO-level rows to prevent target duplication and inflated totals.

Data Modelling



KPI & Analytical Scope

- **Actual Spend** — Actual Unit Price × Ordered Quantity.
- Total POs — Distinct Purchase Order IDs.
- **On-Time Delivery (OTD %)** — On-time delivered shipments ÷ delivered shipments.
- **Average Lead Time** — Order date to actual delivery date.
- **Rejection Rate %** — Rejected quantity ÷ received quantity.
- **Purchase Price Variance (PPV %)** — (Actual Spend − Contract Spend) ÷ Contract Spend.
- **Freight Cost** — Total freight cost.
- **SLA Gap** — Actual OTD % − Carrier SLA Target.

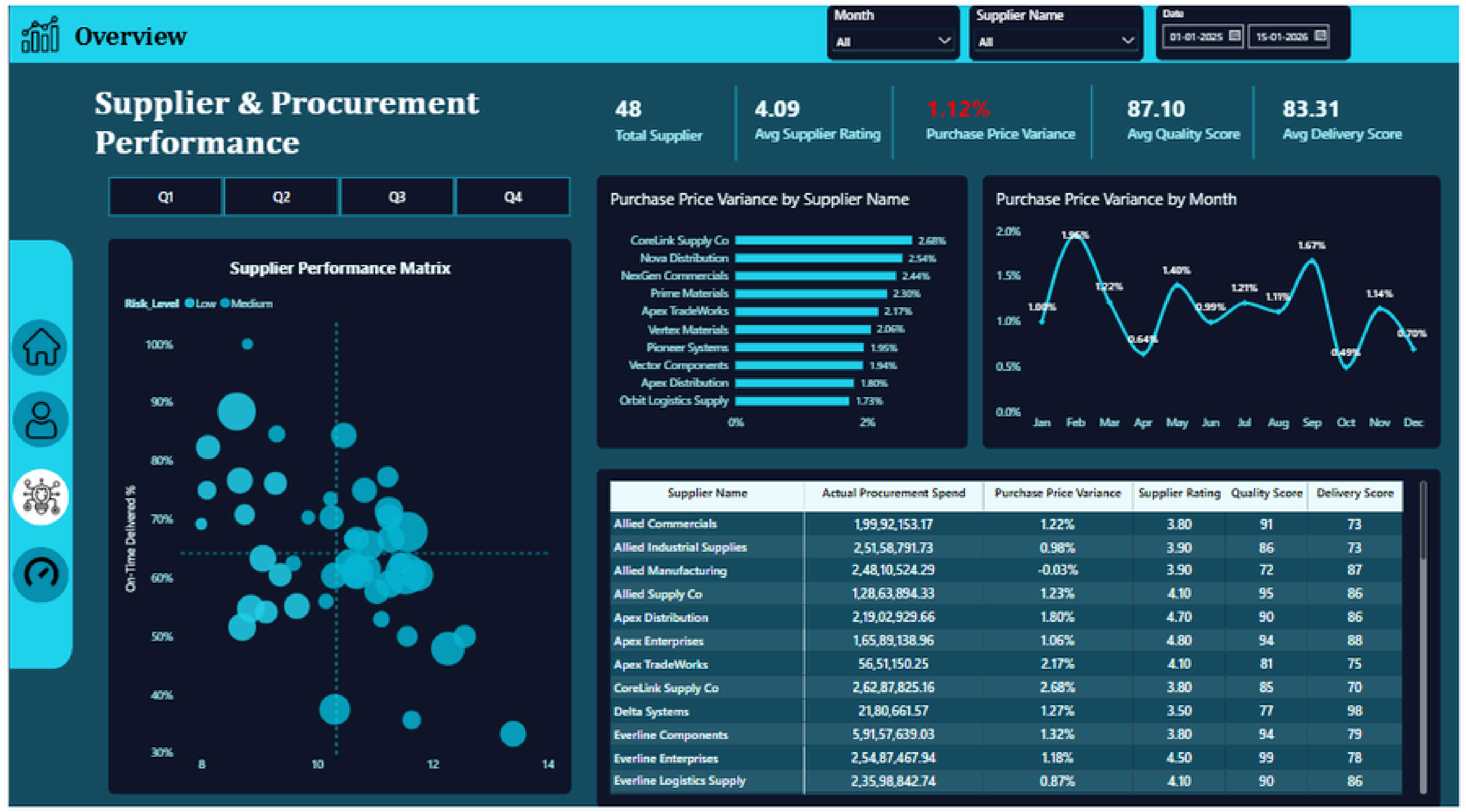
Dashboard Structure

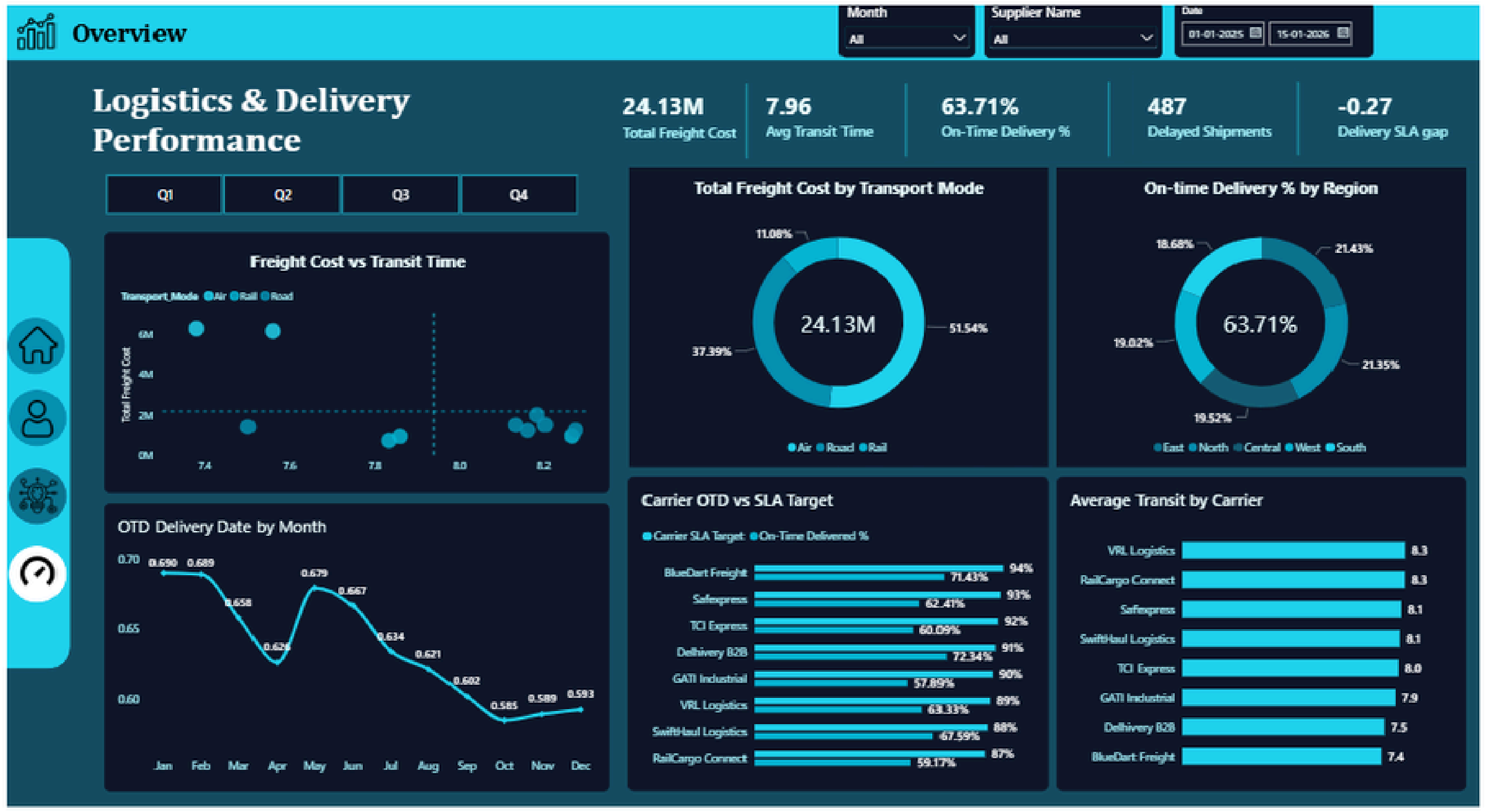
Page 1 — Overview
Provides an executive view of spend, purchase orders, delivery health, quality, supplier risk, and delay root causes, including actual versus target spend, spend variance by category, PO status, supplier risk, and delivery delay analysis.

Page 2 — Supplier & Procurement Performance
Focuses on supplier performance, pricing control, and quality through supplier performance analysis, PPV by supplier, supplier-level detail, and monthly PPV trends.

Page 3 — Logistics & Delivery Performance
Focuses on freight cost, carrier SLA performance, transit speed, and delivery patterns through carrier OTD versus SLA, freight by transport mode, transit time by carrier, and monthly OTD trends.

Dashboard Preview





Key Insights

• **Procurement Spend** - Actual procurement spend stands at approximately ₹991.94M and remains below the target, indicating overall procurement cost control across the analyzed period.

• **On-Time Delivery** - Overall OTD is 63.71%, highlighting a significant opportunity to improve delivery reliability and fulfillment performance.

• **Purchase Orders** - The dashboard tracks 480 purchase orders, providing visibility into overall procurement activity and PO status.

Key Insights

• **Supplier Quality** - The rejection rate is 0.82%, indicating relatively strong supplier quality performance and limited quantity leakage.

• **Lead Time** - Average lead time is 10.43 days, indicating opportunities to further optimize procurement and delivery cycles.

• **Supplier Risk** - Approximately 73% of suppliers fall into the Medium Risk category, making supplier risk monitoring an important management focus.

Key Insights

• **Spend Variance** - Office & IT Supplies show the largest unfavorable spend variance at approximately ₹6.3M,
followed by Electrical & Electronics at ₹3.9M and Maintenance & Tools at ₹2.5M.

• **Delivery Delays** - Road shipments account for the largest share of delayed shipments, with 302 delayed
shipments compared with 113 for Rail and 72 for Air, making road logistics and carrier performance key
improvement areas.

Recommendations

• **Strengthen Road Carrier Performance** - Prioritize root-cause analysis for road shipment delays, review carrier-level SLA performance, and establish corrective actions for consistently underperforming carriers.

• **Improve On-Time Delivery** - Investigate supplier, carrier, region, and transport-mode drivers behind the 63.71% OTD rate and focus improvement efforts on the largest sources of delay.

• **Control Category-Level Spend Variance** - Investigate the high variance in Office & IT Supplies, Electrical & Electronics, and Maintenance & Tools, with emphasis on purchase price, volume, and supplier-level drivers.

Recommendations

• **Proactively Manage Supplier Risk** - Closely monitor the large Medium-Risk supplier segment and combine risk, quality, delivery, and spend exposure when prioritizing supplier reviews.

• **Optimize Lead Time** - Identify suppliers and logistics routes with longer-than-average lead times and evaluate opportunities for process, carrier, and sourcing optimization.

• **Maintain Quality Performance** - Preserve the low rejection rate while continuing supplier quality monitoring to detect emerging quality leakage before it becomes a material cost or service issue.

Business Impact

The dashboard converts raw procurement and logistics records into decision-ready insights for cost control, supplier management, and delivery improvement. It enables stakeholders to move from scattered operational checks toward a unified view of procurement performance, supplier exposure, quality, delivery reliability, and logistics cost pressure

Thank you

Driving Efficiency.
Delivering Value.

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